

CSE2430 - Operating Systems

Homework Introduction

Turn-in Instructions

No hand-in required. Questions are meant for self-directed exam preparation and can be discussed in the midterm Q&A session.

Question 1

Explain the difference between multiprogramming and timesharing when

- a) only a single process is running
- b) two processes are running
- c) more than two processes are running.

Question 2

The kernel abstraction aims at separating privileged operating system code from unprivileged user code. Name three operations that are critical enough so that you would implement them in the kernel and argue why they should be privileged.

Question 3

Educate yourself about the `strace` command (Internet, manpage) and use it to determine how many `open` system calls the execution of the

```
touch .
```

command produces. Explain your findings.

Question 4

Describe an advantage and a disadvantage of modular kernels compared to monolithic kernels.